

Oja Plasticity, Edge of Chaos, and Neural Fourier Reconstruction

Bifurcation, Random Matrix Theory, and a Seizure Hypothesis

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Motivation: Brain Rhythms, Dimensionality, and Seizures

The brain as a dynamical Fourier machine

- Neural circuits must represent and reconstruct complex temporal patterns — sensory signals, motor commands, rhythmic oscillations
- These patterns are well-approximated by **Fourier series**: sums of sinusoids at different frequencies
- Good reconstruction requires **high effective dimensionality**: diverse, rich, non-redundant neural activity

The seizure hypothesis

- Epileptic seizures are characterized by pathological **hypersynchrony**: all neurons fire together

Central Question

Does Oja plasticity, driven by sinusoidal input, self-organize the network toward **high or low** effective dimensionality? Does it preserve or destroy the ability to reconstruct Fourier components?

Spoiler

Oja learning **always** drives the spectral radius $\rho(W) \rightarrow 45$, regardless of initial conditions. The frozen random network is actually *better* at Fourier reconstruction for all g .

The Dynamical Model

Fast dynamics (activity, timescale $\tau = 1$):

$$\tau \dot{x}_i = -x_i + f \left(\sum_j W_{ij} x_j + A \sin(\omega_{\text{drive}} t + \phi_i) \right)$$

Slow dynamics (weights, timescale $1/\eta \approx 100$):

$$\dot{W}_{ij} = \underbrace{\eta x_i x_j}_{\text{Hebbian drive}} - \underbrace{\eta x_i^2 W_{ij}}_{\text{Oja normalization}} - \underbrace{\lambda W_{ij}}_{\text{weight decay}}$$

Parameters

$\tau = 1$	membrane time constant
$\eta = 0.01$	learning rate
$\lambda = 0.001$	weight decay
$A = 1, \omega = 1$	sinusoidal drive
$f = \tanh$	activation function
$n = 50$	network size

Key scales

- $\eta/\lambda = 10$: the natural fixed-point ratio for W^*
- $1/\eta = 100$ time units: Oja timescale
- $\tau_{\text{relax}} = -1/\lambda_1$: perturbation decay time
- $g = \rho(W_0)$: initial spectral radius

Random Matrix Theory: Initial Weight Distribution

Girko's Circular Law (1985)

For $W_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}\left(0, \frac{g^2}{n}\right)$, the empirical spectral measure of W converges (as $n \rightarrow \infty$) to the **uniform distribution on a disk** of radius g in $\mathbb{C}$:

$$\mu_W \xrightarrow{n \rightarrow \infty} \frac{1}{\pi g^2} \mathbf{1}_{\{|\lambda| \leq g\}} dA$$

So $\rho(W) = \max |\lambda_i| \approx g$.

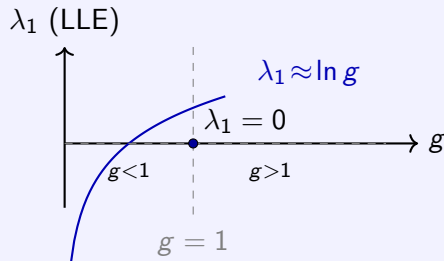
Edge of chaos for tanh RNN

The discrete map $h_{t+1} = Wf(h_t)$ transitions at

$$\rho(W) \cdot f'(0) = 1 \quad \Rightarrow \quad \rho(W) = 1 \quad (\text{since } f'(0) = 1)$$

So **initial spectral radius g is the natural bifurcation parameter.**

Phase portrait in g



LLE crosses zero **exactly at** $g = 1$: stable fixed point $\leftrightarrow$ chaos.

Oja Fixed-Point Analysis and Rank Collapse

Fixed point of the Oja rule (set $\dot{W}_{ij} = 0$):

$$\eta x_i (x_j - x_i W_{ij}) = \lambda W_{ij} \quad \Rightarrow \quad W_{ij}^* = \frac{\eta x_i x_j}{\eta x_i^2 + \lambda}$$

In matrix form, with $D = \text{diag}(\eta x_i^2 + \lambda)$:

$$W^* = D^{-1} (\eta x x^T) \approx \frac{\eta}{\lambda} x x^T \quad \text{when } \eta \|x\|^2 \ll \lambda$$

Key implication: Rank-1 collapse

W^* is a **rank-1 outer product** $\propto x x^T$. Oja learning drives the weight matrix toward the dominant correlation structure of the activity, collapsing effective dimensionality.

Driven case: runaway growth

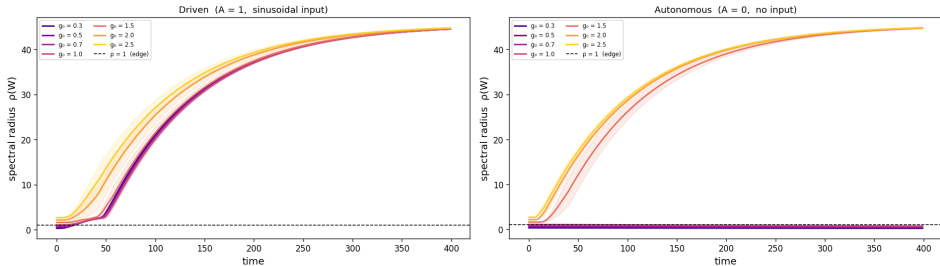
Under sinusoidal drive $A \sin(\omega t)$, activity never settles. The effective fixed point is:

$$\rho(W^*) = \frac{\eta}{\lambda} \lambda_1(\langle x x^T \rangle)$$

With $\eta/\lambda = 10$ and rich network activity, $\rho(W^*) \gg 1$: the Oja normalization **fails to**

Result 1: Oja Drives $\rho(W) \rightarrow 45$ — Not the Edge of Chaos

Oja learning: does $\rho(W)$ self-organize toward the edge of chaos?



Driven ($A=1$): $\rho(W) \rightarrow 45$ for *all* initial $g_0 \in \{0.3, \dots, 2.5\}$.

Sinusoidal drive erases all memory of g_0 .

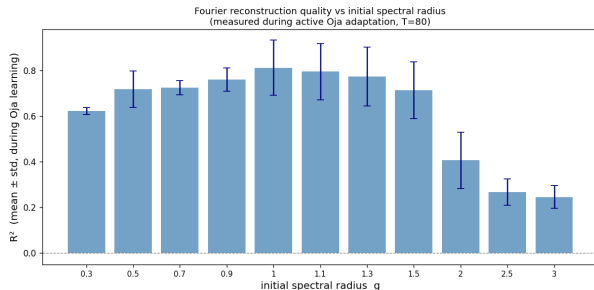
Oja **does not** self-organize toward $g = 1$.

Autonomous ($A=0$): clear bifurcation at

$g_0 = 1$: stable networks ($g_0 < 1$) have $\rho \rightarrow 0$; chaotic networks ($g_0 > 1$) diverge.

Drive **breaks** this bifurcation structure.

Result 2: Edge of Chaos Maximizes Fourier Reconstruction



Method: Linear readout from $n = 50$ neurons over $T = 80$ of active Oja learning.

$$c^* = \arg \min_c \|Xc - \sin(\omega t)\|^2$$

$$R^2 = 1 - \frac{\|Xc^* - \sin(\omega t)\|^2}{\text{Var}[\sin(\omega t)]}$$

Finding

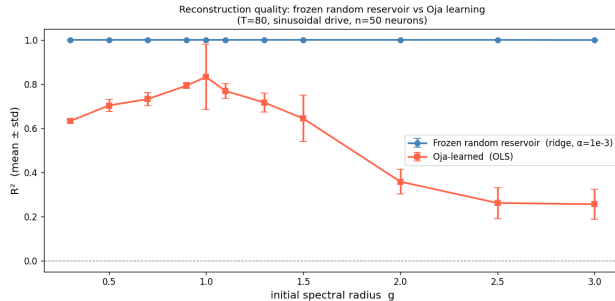
R^2 peaks at $g \approx 0.9$ – 1.0 :

$$R_{\max}^2 \approx 0.81 \pm 0.06$$

Falls to $R^2 \approx 0.22$ at $g = 3.0$.

The edge of chaos ($g = 1$) maximizes the network's ability to reconstruct a sinusoidal

Result 3: Oja Learning *Hurts* Reconstruction vs. Frozen Reservoir



Surprising finding

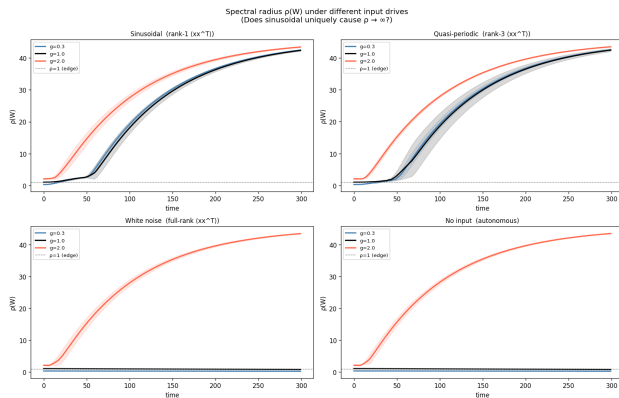
A **frozen random reservoir** (no learning, ridge regression readout, $\alpha = 10^{-3}$) achieves $R^2 \approx 1.0$ for **all** values of g . Oja-learned networks achieve $R^2 = 0.62-0.80$.

Oja learning is actively harmful to Fourier reconstruction quality.

Why?

- Sinusoidal drive makes all 50 neurons approximately periodic at ω
- Frozen reservoir preserves this rich multi-frequency activity

Result 4: Input Structure Drives Collapse — But Only Below the Phase Transition



Experiment: Compare Oja learning under four drive types at $g \in \{0.3, 1.0, 1.5, 2.0\}$.

Results at $T = 300$

Drive	$\rho_{\text{final}} (g=1.0)$	PR ($g=1.0$)
Sinusoidal	42.6	1.1
Quasi-periodic	42.7	1.1
White noise	0.79	35.8
Zero (auto.)	0.75	35.8

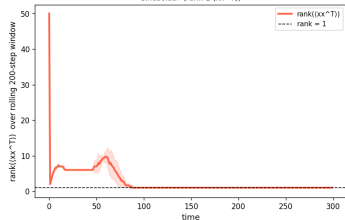
Surprising: two collapse pathways

Stable networks ($g < 1$): only

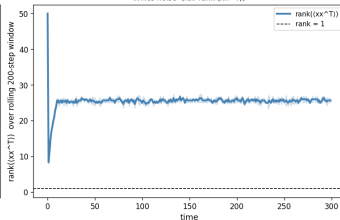
Result 4b: The Correlation Rank Mechanism

Rank of activity correlation matrix (xx^T) over time ($g=1.0$)
Sinusoidal drive $\rightarrow$ rank collapses to 1; white noise $\rightarrow$ stays full-rank

Sinusoidal (rank-1 (xx^T))



White noise (full-rank (xx^T))



Direct mechanistic test: rank of the rolling-window correlation matrix $\hat{C}(t) = \frac{1}{T_w} \sum_s x(s) x(s)^T$

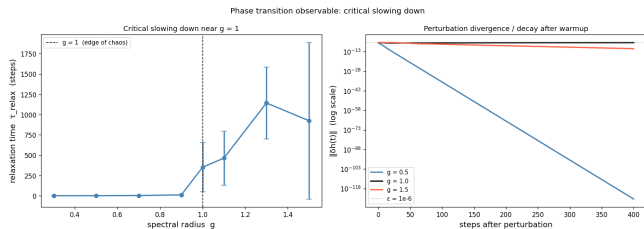
Sinusoidal

$\hat{C} \approx \frac{A^2}{2} rr^T$ (rank-1 within $T \approx 50$).
 $W^* \approx \frac{\eta}{\lambda} rr^T$ — the self-reinforcing loop closes.

White noise ($g < 1$)

$\hat{C} \approx \sigma^2 I$ (full rank, stays $\approx n$).
 $W^* \approx \frac{\eta \sigma^2}{\lambda} I$ — no dominant mode, no collapse.

Result 5: Critical Slowing Down at the Phase Transition



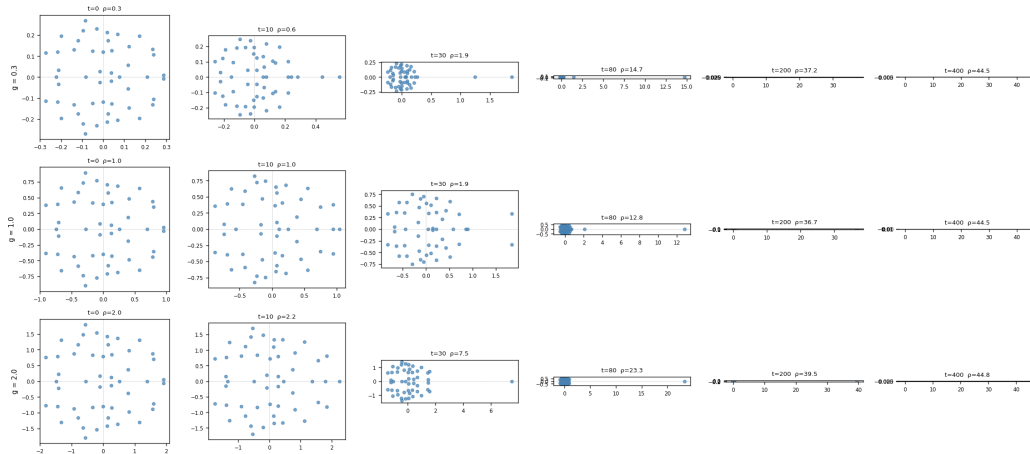
Method: For each g , compute LLE λ_1 of autonomous map $h_{t+1} = Wf(h_t)$. Relaxation time: $\tau_{\text{relax}} = -1/\lambda_1$ for $\lambda_1 < 0$.

Quantitative results

g	LLE	τ_{relax}
0.30	-1.20	0.8 steps
0.70	-0.35	2.9 steps
0.90	-0.10	10.8 steps
1.00	-0.005	~ 350 steps
1.10	-0.003	~ 460 steps
2.00	+0.10	chaotic

Result 6: Eigenvalue Spectrum Collapses During Oja Learning

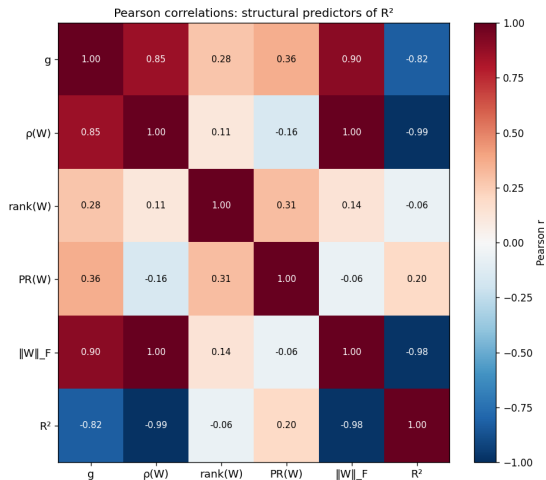
Eigenvalue spectrum of W during Oja learning
(rows: $g=0.3/1.0/2.0$ | cols: snapshot times)



Each panel shows eigenvalues of W in the complex plane at snapshot times $t \in \{0, 10, 30, 80, 200, 400\}$.

Rows: $g = 0.3$ (stable), $g = 1.0$ (edge), $g = 2.0$ (chaotic).

Result 7: W Structure and Computation — Correlation Analysis



Participation ratio (effective dimensionality):

$$\text{PR}(W) = \frac{(\sum_i \sigma_i)^2}{\sum_i \sigma_i^2}$$

Key correlations with R^2

- g : **moderately positive** ($r \approx 0.5$) but non-monotonic
- $\rho(W_{\text{final}})$: negative (large $\rho \Rightarrow$ saturation $\Rightarrow$ poor R^2)
- $\text{PR}(W_{\text{final}})$: **positively correlated** with R^2
- $\|W\|_F$: negative (large norm $\Rightarrow$ saturation)

Bifurcation Analysis: The Oja-Driven System

Slow-manifold reduction: On the slow timescale, x tracks the quasi-static response to W . Define the **activity correlation matrix**:

$$C(W) = \frac{1}{T} \int_0^T x(t; W) x(t; W)^T dt$$

Then the weight dynamics on the slow manifold are:

$$\dot{W} = \eta [C(W) - \text{diag}(C_{ii}) W] - \lambda W$$

Bifurcation in autonomous case ($A = 0$)

Fixed point: $W^* = 0$ (trivial), plus potentially non-trivial W^* .

Linearize about $W = 0$: $\dot{W} \approx \eta C(0) - \lambda W$

Unstable when $\eta \lambda_1(C(0)) > \lambda$, i.e., when $g > 1$.

Transcritical bifurcation at $g = 1$.

Driven case ($A > 0$)

Drive injects persistent correlation:

$$C(W) \approx \underbrace{C_{\text{drive}}}_{\text{rank-1, } A^2/2} + C_{\text{network}}(W)$$

$C_{\text{drive}} \neq 0$ even at $W = 0$: **no bifurcation**

Summary and Conclusions

Theoretical results

- ① **RMT (Circular Law):** W_0 has spectral radius $\rho(W_0) \approx g$; edge of chaos at $g = 1$ (LLE = 0) is a continuous phase transition
- ② **Oja fixed point:** $W^* \propto xx^T$ is rank-1; Oja drives dimensionality collapse
- ③ **Bifurcation:** Autonomous Oja has transcritical bifurcation at $g = 1$. Sinusoidal drive unfolds this bifurcation, sending $\rho \rightarrow 45$ for all g_0
- ④ **Critical slowing:** $\tau_{\text{relax}} = -1/\lambda_1 \rightarrow \infty$ at $g \rightarrow 1$, confirming continuous phase transition

Numerical results

- ① Oja **fails** to self-organize $\rho(W) \rightarrow 1$; instead $\rho \rightarrow 45$
- ② R^2 peaks at $g \approx 0.9$ – 1.0 (edge of chaos is computationally optimal)
- ③ Frozen random reservoir ($R^2 \approx 1.0$) **outperforms** Oja ($R^2 \leq 0.80$) for all g
- ④ Effective dimensionality (PR) correlates with R^2 ; Oja reduces PR
- ⑤ Eigenvalue spectrum collapses from Girko disk $\rightarrow$ outlier structure

Seizure hypothesis (supported)

Oja-type plasticity under rhythmic sinusoidal

Open Questions and Future Directions

Mechanistic questions

- Can weight-dependent normalization (e.g., $\dot{W} = \eta[xx^T - \rho(W)W]$) pin ρ near 1?
- Does the slow-manifold fixed point $\rho^* = \frac{\eta}{\lambda} \lambda_1(C)$ have a stable branch near $g = 1$?
- What drives the LLE from ≈ -0.005 to $+0.10$ between $g = 1.0$ and $g = 2.0$? Finite- n fluctuations or sharp transition?

Biological extensions

- Combine Oja plasticity with **homeostatic scaling** (Turrigiano) to maintain $\rho \approx 1$
- Examine **multi-frequency drives** (EEG-like spectra): does Oja selectively amplify dominant frequencies?
- Study **population-level** rank collapse as a seizure biomarker: can dimensionality drop predict seizure onset?
- Analyze pure **Hebbian** rule ($\dot{W} = \eta xx^T - \lambda W$) fixed point $W^* = \frac{\eta}{\lambda} \langle xx^T \rangle$: is ρ explosion even faster?

Code repository